Computational Method in Chemical Engineering T2730C2

Textbooks:

- Lorens T. Biegler, E. Ignacio Grossmann, Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International.
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation, 2nd Edition, Wiley.
- Robin Smith, Chemical Process: Design and Integration, Wiley.
- James M. Douglas, Conceptual Design of Chemical Processes, McGraw Hill International, 1988.

Unit 6 – Design and Scheduling of Batch Processes

- Objectives
- Introduction
- Design of Batch Process Units
- Design of Reactor-Separator Processes
- Design of Single Product Processing Sequences
- Design of Multi-Product Processing Sequencing

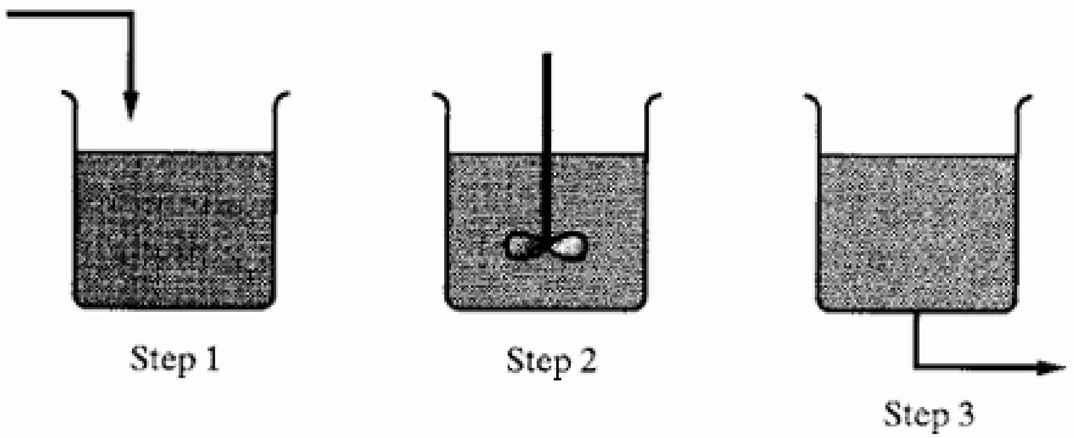
Objectives

- Be knowledgeable about process units executed in batch mode and approaches for optimizing their designs and operations.
- Know how to determine the optimal reaction time for a batch reactor-separate process.
- Be able to schedule recipes for the production of a single chemical product.
- Understand how to schedule batch plants for the production of multiple products.

Introduction

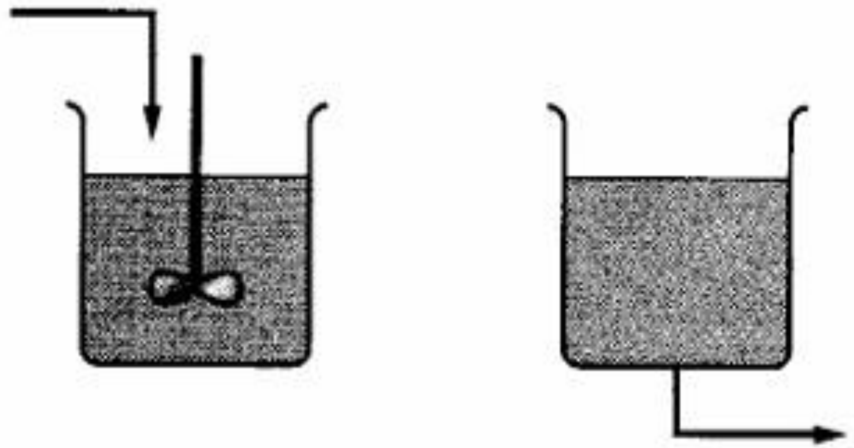
- When **production rates are low**, in the manufacture of specialty chemicals, pharmaceuticals, and electronic materials-it common to **design batch processes or semicontinuous processes** (hybrids between batch and continuous processes).
- In the **batch process** (b), the chemicals are fed before (step 1) and the products are removed after (step 3) the processing (step 2) occurs.
- **Fed-batch processes** (c) combine the first two steps with some or all **chemicals being fed continuously during the processing**. Then, when the processing is finished, the products are **removed batchwise**.
- In **batch-product removal** (d), the **chemicals are fed to the process before processing** begins and **steps 2 and 3 are combined**. The **product is removed continuously** as the processing occurs.
- Fed-batch and batch-product removal processes are semicontinuous processes.

Batch process



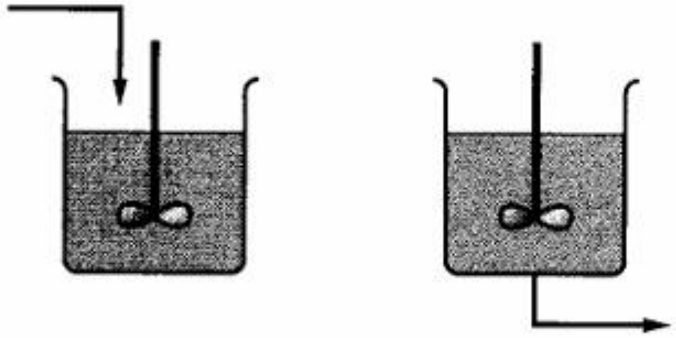
(b)

Fed-Batch process



(c)

Batch-Product removal process



(d)

Introduction

- The challenge in designing a batch, fed-batch, or batch-product removal process is in deciding on **the size of the vessel and the processing time**.
- This is complicated for the latter two processes where the **flow rate and concentration of the feed stream or the flow rate of the product stream as a function of time** strongly influence the performance of the process.
- The determination of optimal operating profiles is referred to as the **solution to the optimal control problem.**
- **Batch and semicontinuous** processes are utilized often when **production rates are small, residence times are large, and product demand is intermittent**, especially when the demand for a chemical is interspersed with the demand for one or more other products and the quantities needed and the timing of the orders are uncertain.
- Even when the demand is continuous and the production rates are sufficiently large to justify continuous processing, batch and semicontinuous processes are often designed to provide a reliable, though inefficient, route to the production of chemicals (exothermic reactions, control issues).

Design of Batch Process Units

- When designing a process unit to operate in batch mode, it is usually desired to **determine the batch time τ, and the size factor S** (volume per unit mass of product), **that maximize an objective** like the amount of product.
- To accomplish this a dynamic model of the process unit is formulated and the degrees of freedom adjusted.
- There are many ways to formulate this optimal control problem.
- To simplify the discussion, **models are presented and studied for various input profiles**, to see how they affect the objectives.
- Emphasis is not placed on the formal methods of optimization.

Batch Processing

- For conventional batch processing, with no material transfer to or from the batch, performance is often improved by **adjusting the operating variables, such as temperature and agitation speed.**
- Through these adjustments, **reactor conversion is improved**, thereby **reducing the batch time to achieve the desired conversion**.

Example 12.1: Exothermic Batch Reactor (Den 1969)

- Consider a batch reactor to carry out the exothermic reversible reaction:

$$n_1 A \rightleftharpoons n_2 B$$

- where the rate of consumption of A is:

$$r\{c_A, c_B, t\} = c_A^{n_1} k_1^{\circ} e^{\frac{-E_1}{RT}} - c_B^{n_2} k_2^{\circ} e^{\frac{-E_2}{RT}}$$

Batch Processing

- and where E1 < *E2* for the exothermic reaction.
- The reaction is charged initially with A and B at concentrations, c_{A0} and c_{B0}.
- To achieve a specified fractional conversion of A, X = $(c_{A0} - c_A)/c_{A0}$, determine the profile of operating temperature in time that gives the minimum batch time.

Solution:

- The minimum batch time, $\mathbf{T_{min}}$ **is achieved by integrating the mass balances**, while adjusting T at each point in time **to give the maximum reaction rate**.

$$\frac{dc_A}{dt} = -r\{c_A, c_B, t\}$$

$$c_B\{t\} = c_{B_o} + \frac{n_1}{n_2}[c_{A_o} - c_A\{t\}]$$

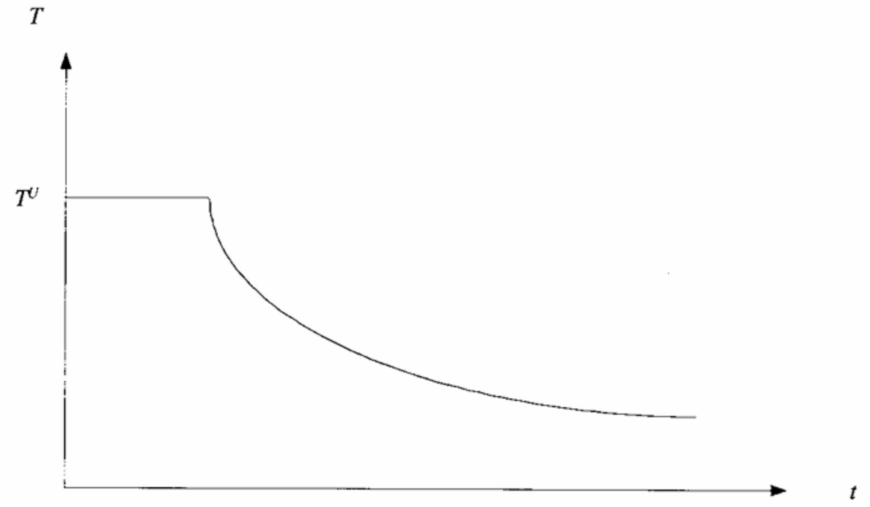
Batch Processing

- The temperature at the maximum reaction rate is obtained by differentiation of r with respect to *T:*

$$\boxed{\frac{\mathrm{d}r}{\mathrm{d}T} = 0}$$

differentiating $$r\{c_A, c_B, t\} = c_A^{n_1} k_1^{\circ} e^{\frac{-E_1}{RT}} - c_B^{n_2} k_2^{\circ} e^{\frac{-E_2}{RT}}$$

$$\boxed{T_{\mathrm{opt}} = \frac{E_2 - E_1}{R \ln \dfrac{c_B^{n_2} k_2^{\circ} E_2}{c_A^{n_1} k_1^{\circ} E_1}}}$$



- Upper bound in temperature is T^U
- When $T_{opt} > T^U$, the reactor temperature is adjusted to the upper bound, T^U.
- Then, as conversion increases, the reactor temperature decreases, leveling off to the equilibrium conversion.

Fed-Batch and Batch-Product Removal Processing

Fed-Batch Processing:

- Fermentation processes for the production of drugs are usually carried out in fed-batch reactors.
- In these reactors, it is desirable to find **the best profile for feeding substrate** into the fermenting broth.

Batch-Product Removal processing:

- When distillations are carried out in batch mode, the still is charged with the feed mixture and the heat is turned on in the reboiler.
- The lightest species concentrate in the distillate, which is condensed and recovered in batch-product removal mode.
- As the light species is recovered, it is accompanied by increasing fractions of the heavier species unless a strategy is applied to maintain a high concentration of light species.
- **For multicomponent separations, to simplify operation it is often satisfactory to adjust the reflux rate once or twice while recovering each species.**
- The reflux rate is increased as the difficulty of the separation between the light and heavy key components increases.

Design of Reactor-Separator Processes

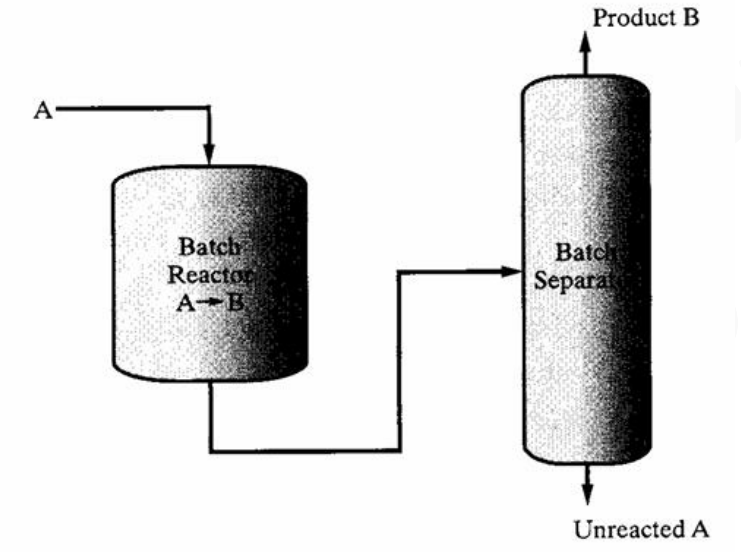
- An approach to solving the **optimal control problem is introduced for reactor-separator** processes.
- The approach involves the **simultaneous determination of the batch times and size factors for both of the process units.**
- Furthermore, the interplay between the two units involves trade-offs between them that are adjusted in the optimization.

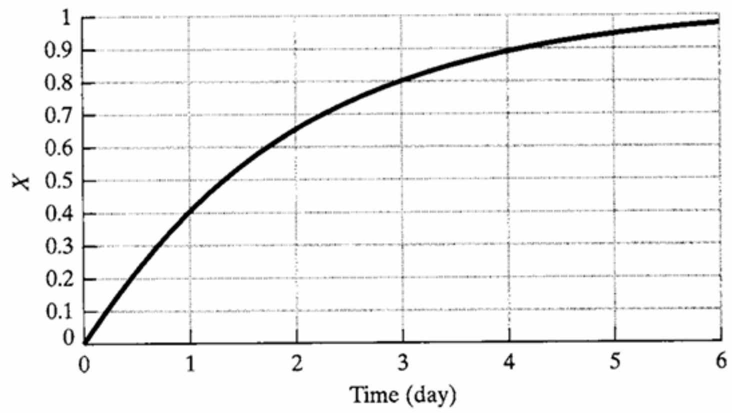
Example 12.4:

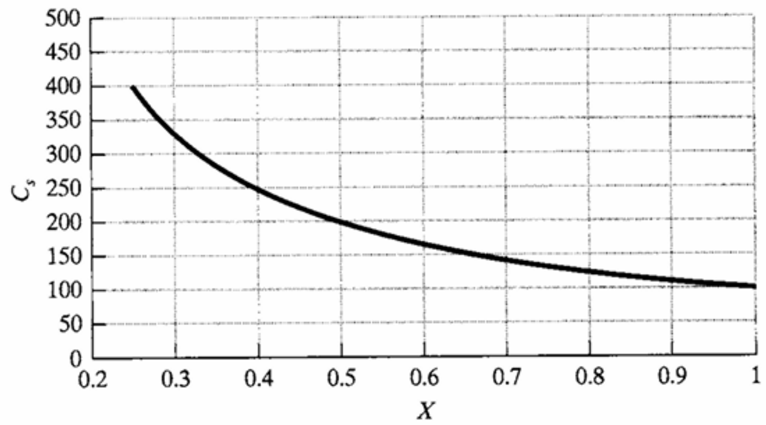
Consider the batch reactor–separator combination in Figure 12.6, initially presented by Rudd and Watson (1968). In the reactor, the isothermal, irreversible reaction, A $\rightarrow$ B, is carried out. The separator recovers the product B from unreacted A. The rate constant for the reaction is $k = 0.534$ day^{-1}. It is assumed that the cost of A is negligible compared to the value of the product, B, $C_B =$ \$2.00/lb, that the operating costs of the reactor are proportional to the batch time (that is, $C_O = \alpha t$,

Design of Reactor-Separator Processes

where α = \$100/day), and that the cost of separation per batch, C_S, is inversely proportional to the conversion of A in the reactor, X (that is, $C_S = KV/X$), where K is the proportionality constant and V is the holdup volume in the reactor. Find the batch time for the reactor when the gross profit in \$/day is maximized. Let c_{A_o} = 10 lb/ft^3, V = 100 ft^3, and K = 1.0 \$/(ft^3·batch). Note that as the conversion of A to product B increases, the cost of separation decreases, as shown in Figure 12.7b.







Design of Reactor-Separator Processes

- For the first-order reaction

$$X = 1 - e^{-kt} \qquad c_B = c_{Ao}X$$

- It is desired to maximize the gross profit, GP, in $/day

$$\max_t \text{GP} = \frac{1}{t}\left[C_B V c_B\{t\} - C_O\{t\} - C_S\{X\{t\}\}\right]$$

- To locate the maximum, substitute the equations above and differentiate: $\boxed{\frac{d\text{GP}}{dt} = 0}$

- At **maximum, reactor batch time is τ_{opt} = 1.35 day, with a fractional conversion, X_{opt}> of 0.514**.
- At longer times, while the revenues increase due to increased conversion and the separation costs decrease, the gross profit decreases due to the increase in the batch.
- At shorter times, the revenues decrease due to smaller conversion and separation costs increase due to a more difficult separation.
- At the maximum, GP_{opt} = $516.8/day. Note that GP < 0 when t < 0.507 day.

Design of Reactor-Separator Processes

- Example 12.5

Figure 12.8 shows an isothermal batch reactor, in which the irreversible reactions, A → B → C, take place, with B the desired product and C an undesired byproduct. The reactions are irreversible and first-order in the reactants. The reaction rate constants are $k_1 = 0.628\ \text{hr}^{-1}$ and $k_2 = 0.314\ \text{hr}^{-1}$. The reactor products are fed from an intermediate storage tank (not shown) to a batch distillation column, from which the most volatile species, A, is removed first in the distillate, followed by the product B, which is the intermediate boiler. For specified reactor volume, V_r, and column volume, V_c, it is desired to determine the batch times for the reactor and column that minimize the cost of producing the desired amount of product B, B_{tot} moles, in a single campaign. Following Barrera and Evans (1989), the analysis is simplified by assuming that the distillation column produces pure A until it is depleted from the still, followed by pure B. Furthermore, specifications include the total molar concentration in the reactor, C, the distillate flow rate, F_d, the time horizon within which the campaign must be completed, τ_{hor}, and the cleaning times between batches for the reactor and distillation column, t_{cr} and t_{cc}. In addition, several cost coefficients are specified, including the cost of fresh feed, P_A (\$/mol), recycling credit for A, P_{rA}, cost or credit for the byproduct C, P_C; the rental rates of the reactor, r_r (\$/hr), intermediate storage, r_s, and distillation column, r_c; the costs of cleaning the reactor, C_{clr} (\$/batch), and distillation column, C_{clc}; and the utility cost per mole of distillate, P_u. These specifications are given in Table 12.1.

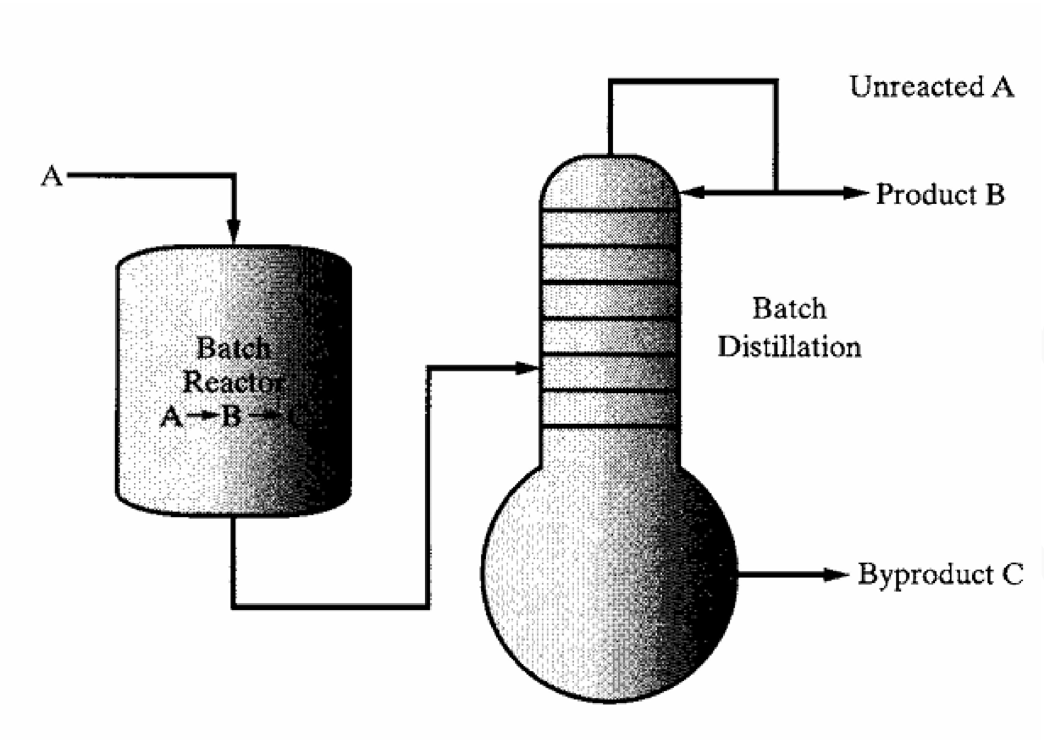
By integration of the kinetic rate equations, mole fraction profiles obtained

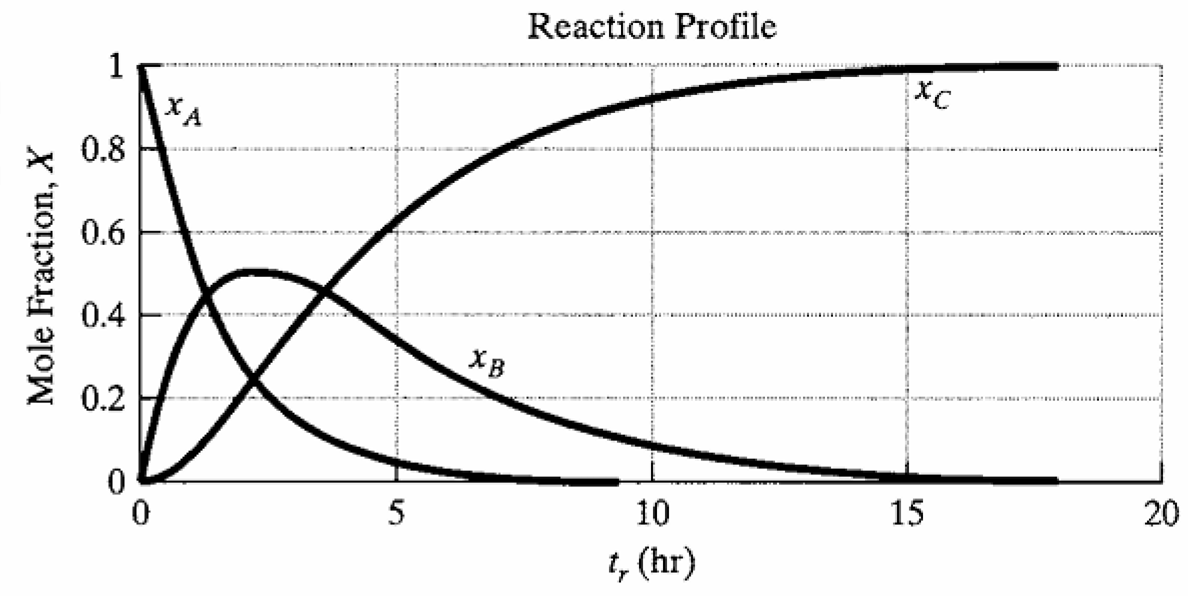
$$x_A = e^{-k_1 t_r}$$

$$x_B = \frac{k_1 (e^{-k_1 t_r} - e^{-k_2 t_r})}{k_2 - k_1}$$

$$x_C = 1 - x_A - x_B$$

$k_1 = 0.628\ hr^{-1}$ and $k_2 = 0.314\ hr^{-1}$





The minimization of the campaign costs is expressed as:

$$\min_{t_r, \tau_c} \phi = \phi_{rm} + \phi_{eq} + \phi_{cl} + \phi_u$$

where t_r is the reaction time (reactor operating time)
τ_c is the batch time for the distillation column

the campaign costs for the raw materials, equipment rental, cleaning, and utilities

$$\phi_{rm} = CV_r[P_A + P_{rA}x_A + P_C x_C]\tau_{tot}/\tau_r$$

$$\phi_{eq} = [r_r + r_s + r_c]\tau_{tot}$$

$$\phi_{cl} = [C_{clr}/\tau_r + C_{clc}/\tau_c]\tau_{tot}$$

$$\phi_u = P_u F_d t_c \tau_{tot}/\tau_c$$

Table 12.1 Specifications for Example 12.5

$V_r = 3{,}503$ L, $V_c = 784$ L
$B_{tot} = 800$ mol
$C = 0.5$ mol/L
$F_d = 206.2$ mol/hr
$\tau_{hor} \leq 24$ hr
$t_{cr} = 0.8$ hr, $t_{cc} = 0.5$ hr
$P_A = 1$ \$/mol, $P_{rA} = 0.2$ \$/mol, $P_C = 0.4$ \$/mol
$r_r = 100$ \$/hr, $r_s = 20$ \$/hr, $r_c = 100$ \$/hr
$C_{clr} = 100$ \$/batch, $C_{clc} = 100$ \$/batch
$P_u = 0.45$ \$/mol

tc/τ_c is the fraction of column batch time during which the column is operational (and using utilities), τ_{total} is the total batch time.
These quantities are multiplied to give Φ_U, the cost of utilities per batch.
This assumes that the utilities are applied during a tc/τ_c fraction of the reactor operation.

- The minimization is carried out subject to a mass balance for the reactor: $B_{tot} = CV_r x_B \tau_{tot}/\tau_r$

- Equation that assures periodicity in the storage tank (equal volumes processed per unit time): $\frac{V_r}{\tau_r} = \frac{V_c}{\tau_c}$

- Mass balance for the distillation column $t_c = V_r C(1 - x_C)/F_d$

- **The total campaign time, where τ_r is the batch time for the reactor and t_c is the column operating time**

$$\boxed{\tau_{tot} = \tau_r + \tau_c}$$

Inequality constraints:

$$\tau_{tot} \le \tau_{hor}$$

$$\tau_r - (t_r + t_{cr}) \ge 0$$

$$\tau_c - (t_c + t_{cc}) \ge 0$$

- Solution was obtained using **successive quadratic programming (SQP) in GAMS.**
- The reactor operates for t_r = 4.44 hr
- Product x_A = 0.062, x_B = 0.373, and x_C = 0.565
- **For periodic cycling, its batch time, τ_r = 18.74 hr, which exceeds the total of the reactions and cleaning times.**
- Distillation column operates for t_c = 3.69 hr, cleaning time, t_{cc} = 0.5 hr, batch time for the column, τ_c = 4.19 hr.
- The total time, or cycle time, is τ_{tot} = 22.93 hr, which falls within the horizon time specified, T_{hor} = 24 hr.
- These times correspond to the minimum cost, Φ = $10,240.
- **By varying t_r, it can be shown that this is the minimum cost.**

Design of Single Product Processing Sequences

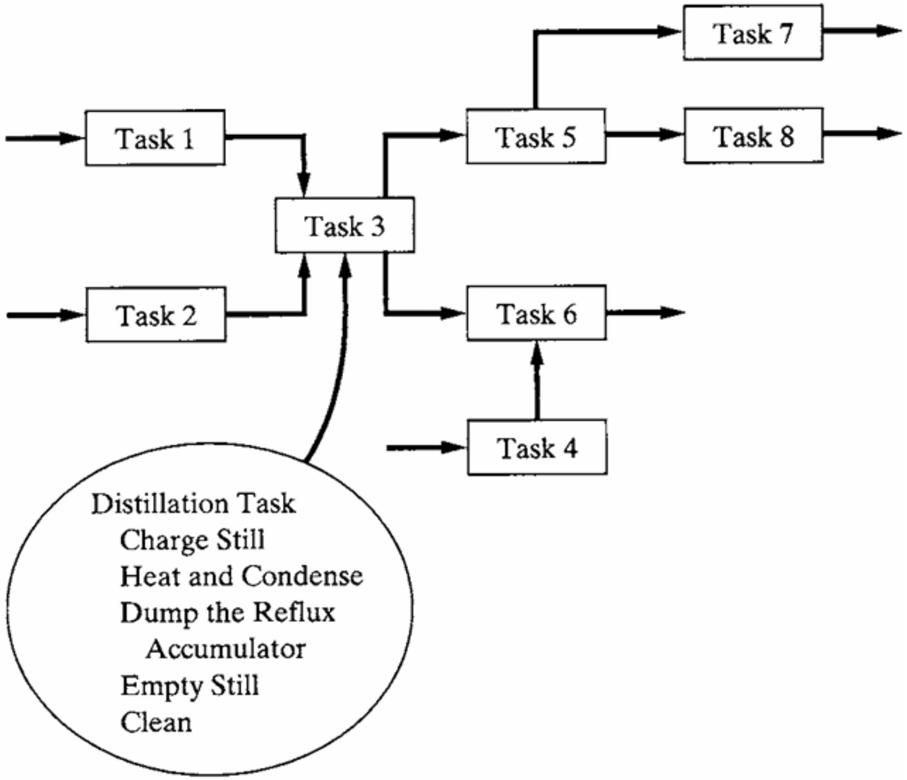
- **The determination of optimal batch times, given batch sizes expressed as batch volumes per unit mass of product, can be demanding computationally.**
- Since most processes, in practice, have recipes with numerous tasks and a comparable number of processing units, it is normally not practical to optimize the batch times for the individual processing units when preparing a schedule of tasks and equipment items for the manufacture of a product.
- **Consequently, when preparing a schedule of tasks and equipment items, it is common to specify batch times for tasks to be performed in specific units, usually with batch sizes, and to optimize cycle times for a specific recipe.**
- In some cases, using the rates of production and yields, the vessel sizes are determined to minimize the cost of the plant while determining the cycle times for a specific recipe.
- ***Batch process design begins with the specification of a recipe of tasks to produce a product.***

Design of Single Product Processing Sequences

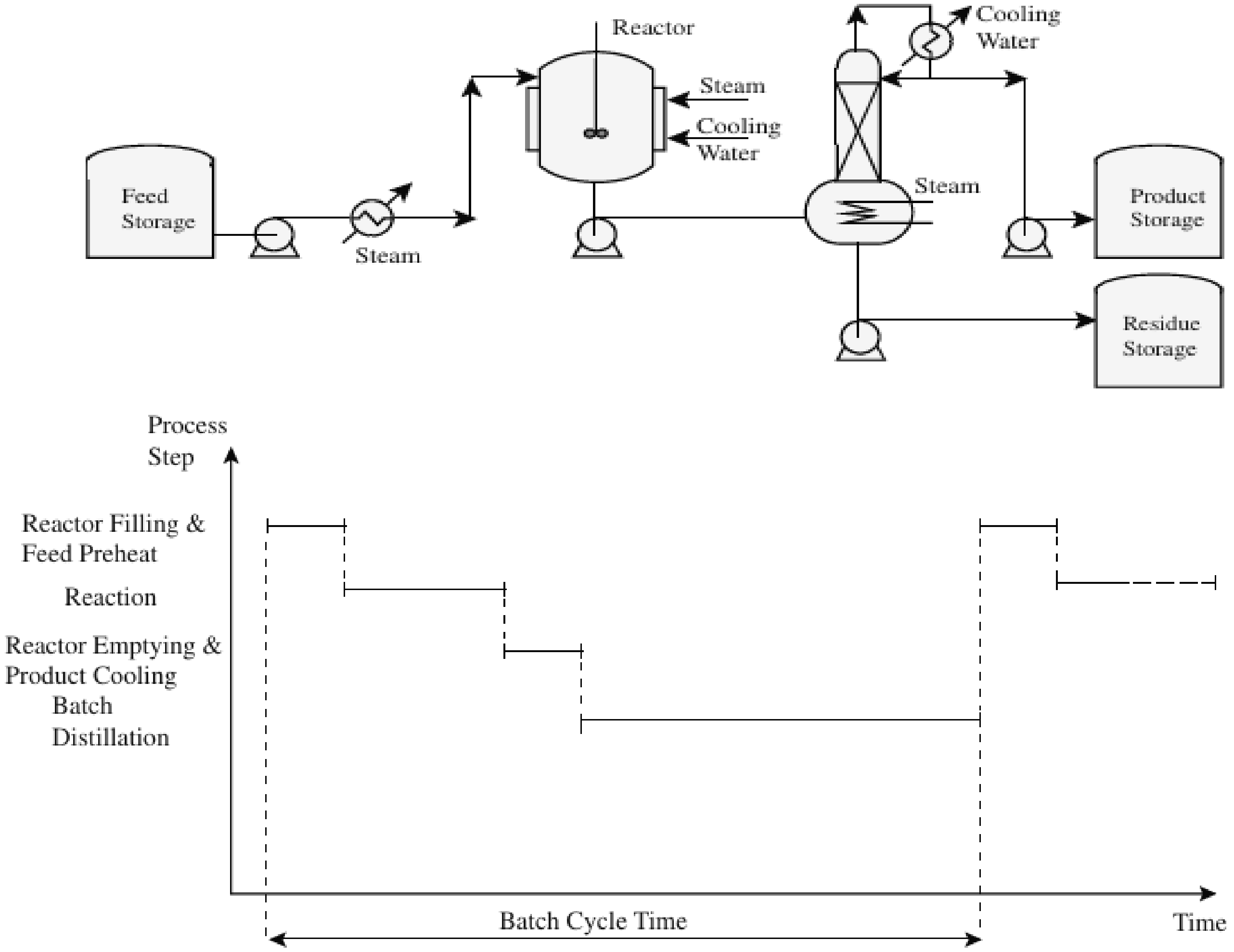
- In batch processes, the tasks are assigned to equipment items, but over specific intervals of time, which vary with batch size, which is often determined by the available equip-ment sizes.

Cycle times:

- In batch processes, consist of a sequence of steps to be carried out in the same equipment unit.
- **Cycle time is the time between the completions of batches.**
- Each step involves a batch time, which is determined by the processing rates and the batch size (amount of the final product manufactured in one batch).
- A production line is a set of equipment items assigned to the tasks in a recipe to produce a product.



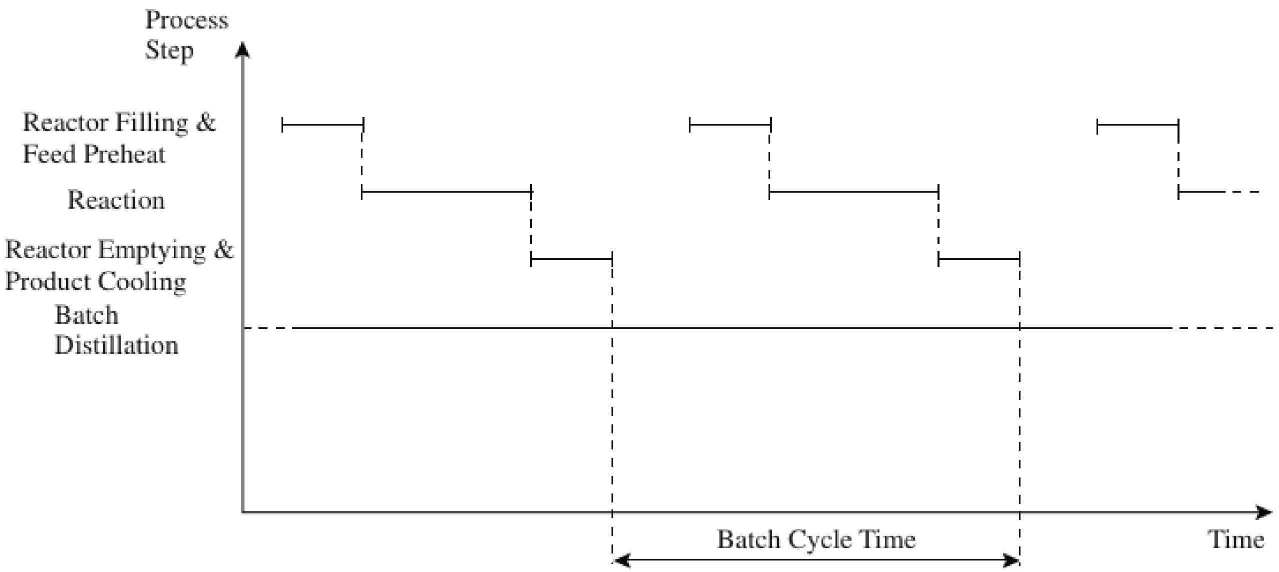
Gantt chart for a simple batch process



- **Gantt or time event chart**
- **Batch cycle time** is the time interval between successive batches of product being produced.
- **Dead time:** considerable periods over which the equipment is **standing idle**, poor utilization of equipment
- **High utilization of equipment is one of the goals of batch process design.**
- This can be achieved by overlapping batches.

Robin Smith, Chemical Process: Design and Integration, Wiley.

Gantt chart for a simple batch process



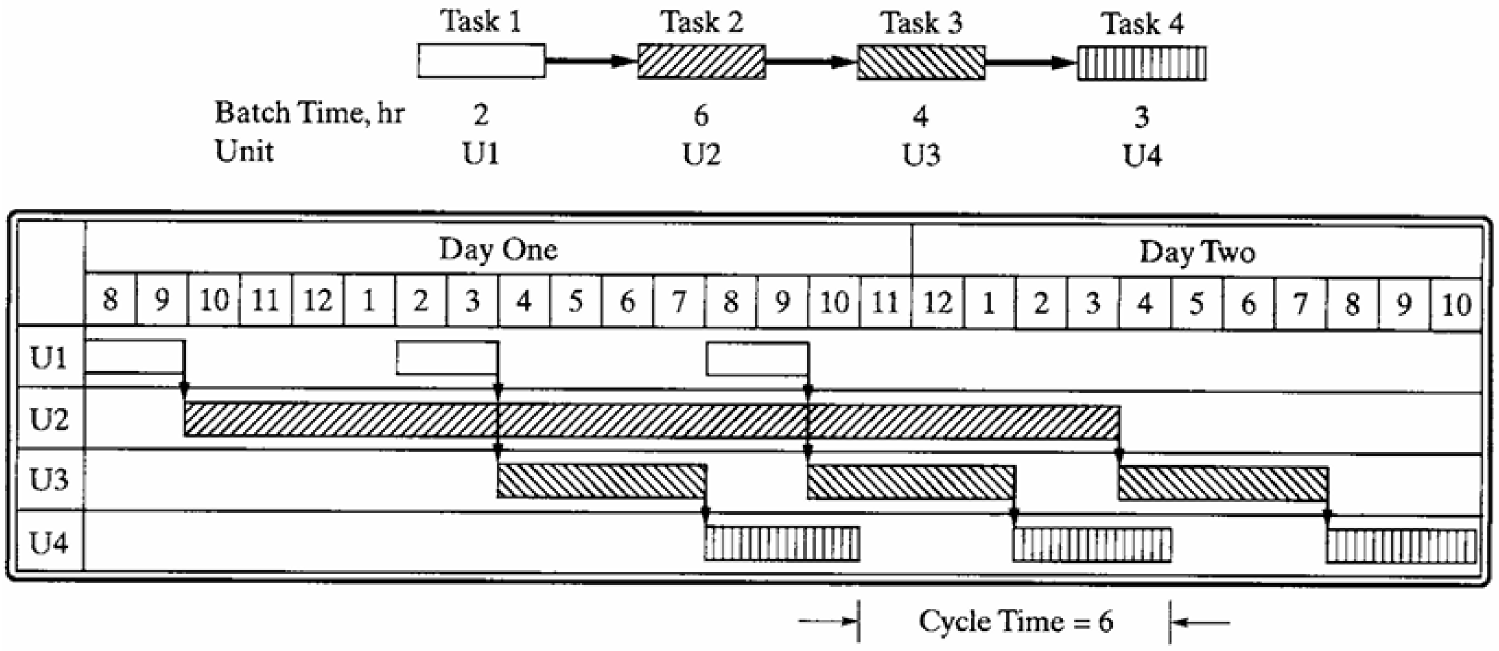
Overlapping batches allows the batch cycle time to be decreased.

- This allows the batch cycle time - the time interval between producing successive batches of product, to be decreased considerably.
- **The step with the longest time limits the cycle time.**
- Alternatively, if more than one step is carried out in the same equipment, the cycle time is limited by the longest series of steps in the same equipment.
- The batch cycle time must be at least as long as the longest step.
- The rest of the equipment other than the limiting step is then idle for some fraction of the batch cycle.

Robin Smith, Chemical Process: Design and Integration, Wiley.

Design of Single Product Processing Sequences– Gantt chart

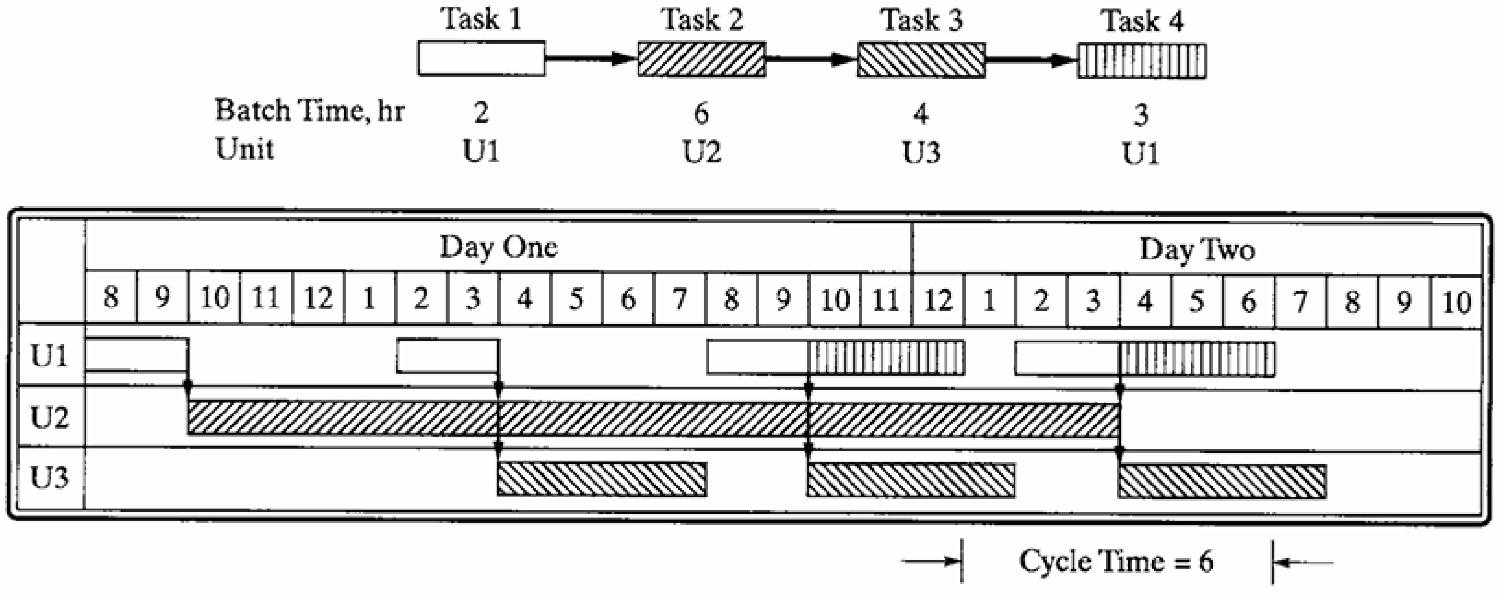
- **To better visualize the schedule of production, an equipment occupation diagram, known as a Gantt chart, is prepared, showing the periods of time during which each equipment item is utilized.**



(a) Distinct Task Assigned to Each Unit

- ***The unit having the longest batch time (6 hr), U2, is the bottleneck unit, as it is always in operation***.
- The batches are transferred from unit-to-unit immediately , called **zero-wait strategy**, with no intermediate storage utilized.
- The cycle time, 6 hr, is the batch time of U2.

Design of Single Product Processing Sequences–Gantt chart

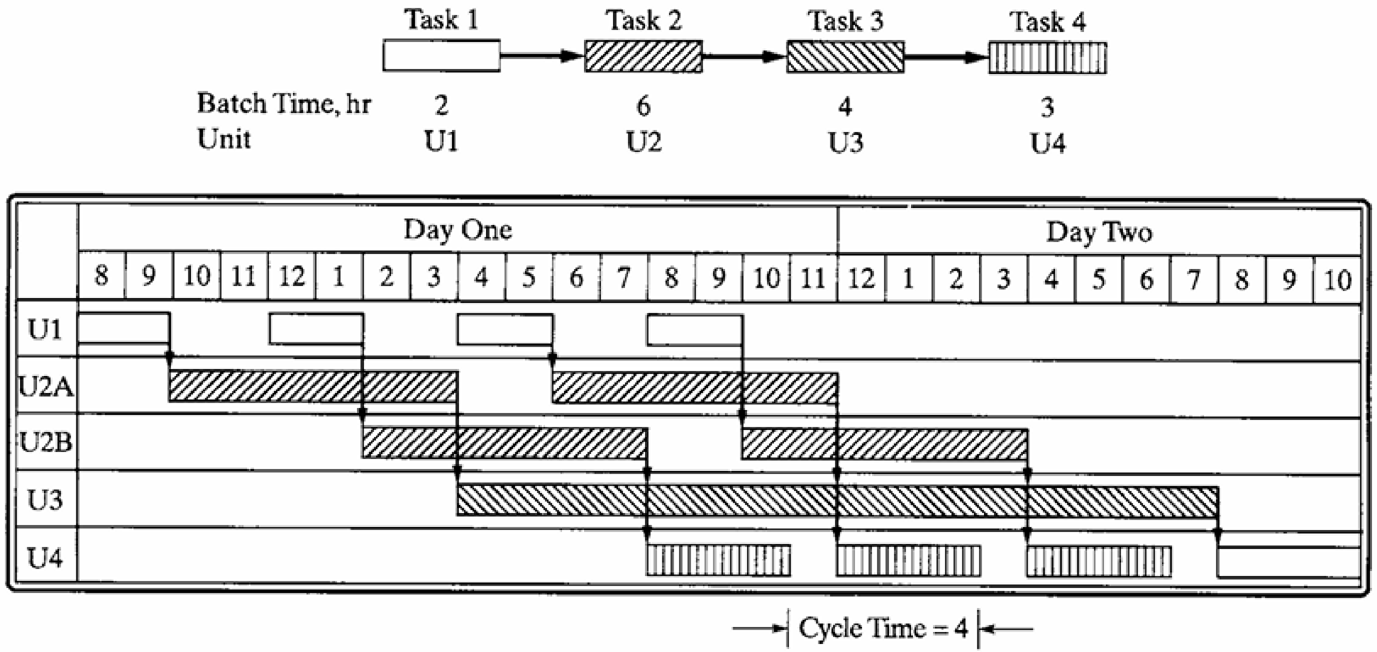


(b) Multiple Tasks Assigned to Same Unit

Multiple tasks assigned to same unit:

- **When the fourth task can be carried out in U1, this unit is better utilized and U4 can be released for production** elsewhere in the batch plant.
- to achieve this schedule, without adding intermediate storage, it is necessary to retain the batch within U3 until U1 becomes available.

Design of Single Product Processing Sequences– Gantt chart



(c) Parallel Units

Parallel units:

- To **increase the efficiency of the schedule and reduce the cycle time**, it is common to add one or more units in parallel.
- When in phase, the batch time for the unit is reduced to τ_j/n_j, where n_j is the number of units in parallel for task j.
- The parallel units can be sequenced out-of-phase, without altering their batch time.
- **The U2 bottleneck is eliminated and the cycle time is reduced to 4 hr.**

Design of Single Product Processing Sequences– Gantt chart

- The batch cycle time, C_T *without* parallel operation, is the maximum of the batch times, τ_j, j = 1, . . . , M, where *M* is the number of unique equipment units.

$$\boxed{CT = \max_{j=1,\ldots,M} \tau_j}$$

- With *nj* units in parallel and in phase, the cycle time is given by:

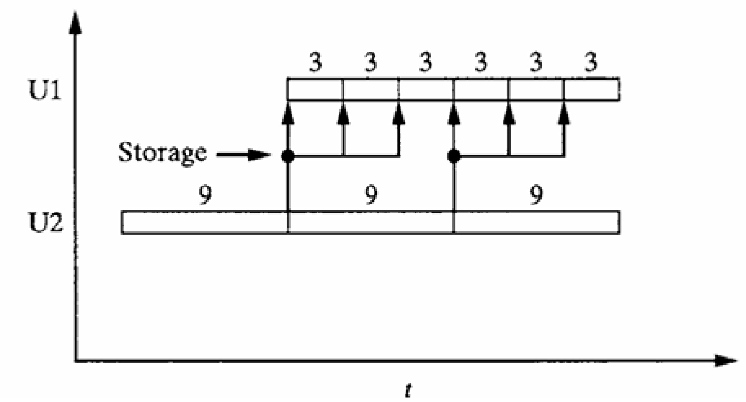
$$\boxed{CT = \max_{j=1,\ldots,M} \frac{\tau_j}{n_j}}$$

- Returning to the example, when two units U2 are installed in parallel to perform task 2

$$CT = \max_{j=1,\ldots,M} \frac{\tau_j}{n_j} = \max\{2, \tfrac{6}{2}, 4, 3\} = 4 \text{ hr}$$

Design of Single Product Processing Sequences – Intermediate Storage

- **Zero-wait (ZW) strategy:** contents of each unit transferred immediately to the next unit, experiencing no delay after its task has been completed
- **Intermediate storage (IS) strategy:** a zero-wait strategy is implemented, with some intermediate storage when necessary
- **Unlimited intermediate storage (UIS):** sufficient to hold the contents of the products from a unit having a lengthy batch time, to be used repeatedly in a unit having half the batch time or less, can greatly reduce cycle time.



Gantt chart with unlimited intermediate storage (UIS)

Design of Single Product Processing Sequences–Batch Size

- Convenient to define the **size factor**, S_j for task *j,* as the capacity required per unit of product.
- **Defined as the volume required to produce a unit mass of product.**
- Size factors can be computed for each task in a recipe.
- Normally, equipment vessel sizes are selected that exceed batch volume by 10 to 20%.
- The batch factor in volume/mass produced is determined by the rate of processing the batch (e.g., kg/hr) multiplied by the batch time (hr) and divided by the density of the batch (kg/L) and the mass of product produced (kg).

Design of Multi-Product Processing Sequencing

- **Multiproduct batch plant:** produces a set of products whose recipe structures are the same, or nearly identical.
- In these plants, each product is produced in the same production line, with multiple processing tasks carried out using the same equipment items.
- Multiproduct batch plants are used for larger volume products having similar recipes, as is often the case for plants that produce a family of grades of a specific product.

- **Multipurpose batch plant:** tasks for products A and B differ in equipment utilized.
- Plants are more flexible and effective for a large number of products that are produced in small volumes, where their vessels are cleaned easily and the presence of trace contaminants in the products is not a concern.
- Their equipment items are utilized more completely, without the idle-time gaps in plants with cyclic campaigns for each product.

Scheduling and Designing Multiproduct Plants

For an existing plant, the scheduling problem involves a specification of the:

(1) product orders and recipes,

(2) number and capacity of the equipment items,

(3) a listing of the equipment items available for each task,

(4) limitations on the shared resources (e.g., involving the usage of utilities and manpower),

(5) restrictions on the use of equipment due to operating or safety considerations.

- In solving the problem to determine an optimal schedule, the order in which tasks use the equipment and resources is determined.
- With specific timings of the tasks provided, that optimize the plant performance (to maximize the gross profit, etc.).

Multiproduct plant

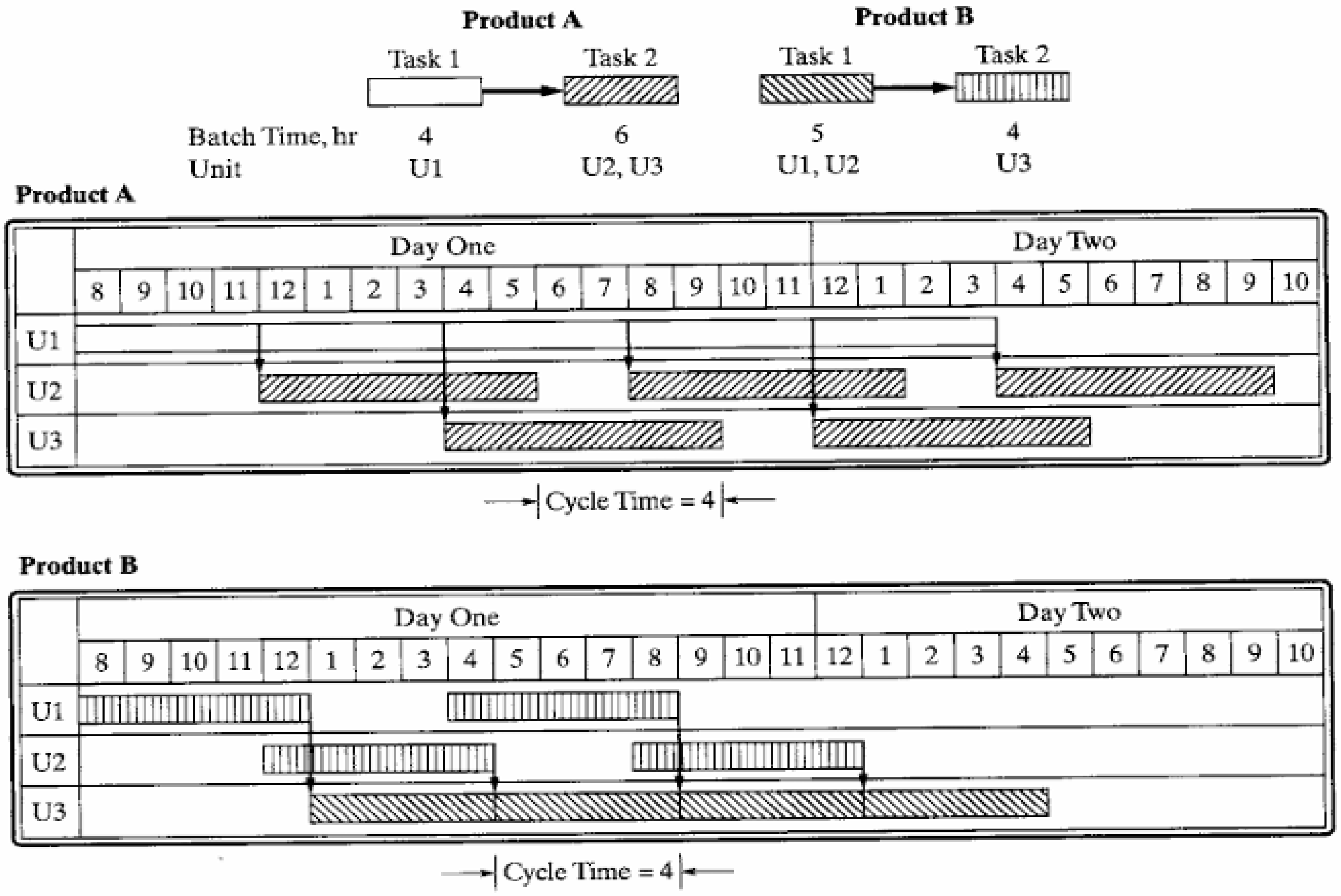


Figure 12.13 Gantt charts for a multipurpose plant.

Multipurpose plant

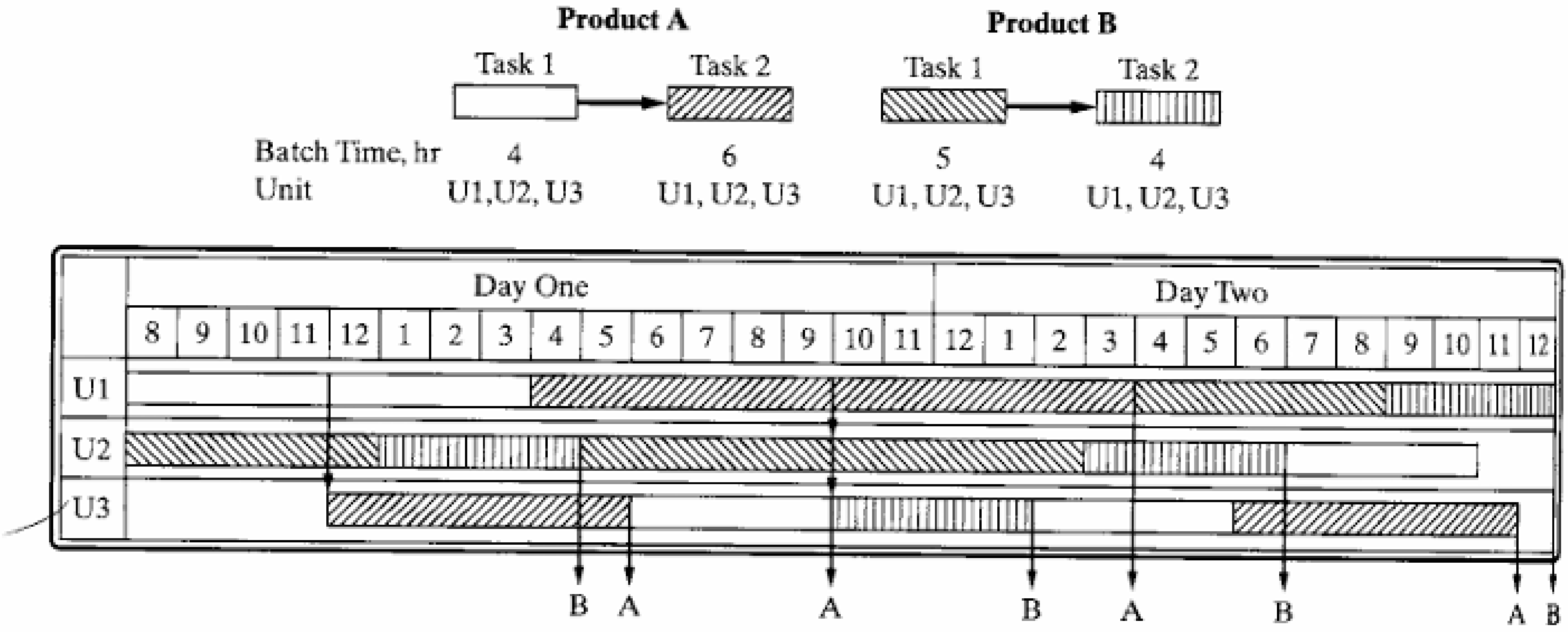


Figure 12.14 Gantt chart for a general multipurpose plant.

Scheduling and Designing Multiproduct Plants

- To formulate the design problem for a multiproduct batch plant involving the processing of batch campaigns in series (one-at-a-time--commonly referred to as a Flowshop plant) as a **mixed-integer nonlinear program (MINLP).**
- It begins with the objective, that is, to **minimize the total investment cost, C**, where m_j is the number of out-of-phase units assigned to task j (integer variables), M is the number of tasks, and V_j is the size of the unit assigned to task j (usually in L; a_j and α_j are cost coefficients).

$$\boxed{\min C = \sum_{j=1}^{M} m_j a_j V_j^{\alpha_j}}$$

- This objective is minimized commonly subject to inequalities that involve the vessel size, where B_i is the batch size of product *i* (the final product size, typically in kg), and S_{ij} is the size factor for task *j* in producing product *i* (typically in L/kg).

$$V_j \geq B_i S_{ij} \qquad V_j^L \leq V_j \leq V_j^U$$

Quick Recap

- What are the types of batch processes?
- What are the key challenges in designing of batch processes?
- What is the optimal control problem for designing of batch processes?
- Define batch cycle time. Write formula for simple and parallel batch design.
- What is Gantt chart? Draw a Gantt chart for simple batch process.
- What are the storage types?
- What are multiproduct and multipurpose batch processes?
- Write steps in designing specifications for scheduling tasks in multiproduct processes.